Report: Analysis of DNA/Protein Sequences

CodeAlpha Internship, Bioinformatics, Task 1

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Introduction

Cassava Mosaic Disease (CMD) and Cassava Brown Streak Disease (CBSD) are two viral diseases that destroy approximately 30–40% and up to 70% of cassava harvests in Africa, respectively. The geminiviruses responsible for CMD consist of several species of circular, single-stranded DNA viruses belonging to the genus Begomovirus (Chikoti et Tembo 2022). The capsid forms the virus's outer shell, protects its DNA, facilitates transmission via whiteflies, and subsequently suppresses the host's defenses. The objective of this analysis is to identify proteins homologous to the capsid in order to assess its level of conservation.

1-Sequence Studied

The sequence under study is the capsid protein of the African Cassava Mosaic Virus (ACMV), isolate West Kenyan 844. It was downloaded in FASTA format from UniProtKB on October 13, 2026. The entry has been reviewed by curators (Swiss-Prot), and the protein's existence has been confirmed at the protein level. It consists of 258 amino acids. Table 1 summarizes these characteristics, and Figure 1 displays the UniProt entry. Table 1: Sequence characteristics

Table 1: Sequence characteristics

Characteristic	Value
UniProtKB accession	P03561 (CAPSD_CLVK)
Protein name	Capsid protein (coat protein, CP)
Gene	AR1 (AV1)
Organism	African cassava mosaic virus (isolate West Kenyan 844), ACMV
Taxonomic identifier	10818
Length	258 amino acids
Entry status	Swiss-Prot (reviewed)
Evidence for existence	Protein level

Known 3D structure	PDB 6EK5 (cryo-electron microscopy, 4.2 Å)

Source UniProtKB, accessed October 13, 2026



Figure 1: UniProtKB entry P03561: ACMV capsid protein (West Kenyan isolate 844)

2-Methodology

Homologous sequences were searched for using BLASTp on the NCBI website. Sequence P03561 was compared against the "Non-redundant protein sequences (nr)" database. The search was conducted on October 13, 2026 (RID: CX49UVJJ014). Default settings were used: BLOSUM62 matrix, composition-based statistics, an E-value threshold of 0.05, and a maximum of 100 target sequences. The results were exported in CSV format. The alignment with the isolate from Nigeria (CAD20824.1) was then examined in detail.

3- Results

3-1 Overview

The search yielded 100 sequences. In the summary chart, the hits show an alignment score greater than or equal to 200 and cover all 258 amino acids of the query sequence (Figure 2). BLAST also identified the conserved "Gemini_coat" domain, spanning from approximately position 18 to the end of the protein.

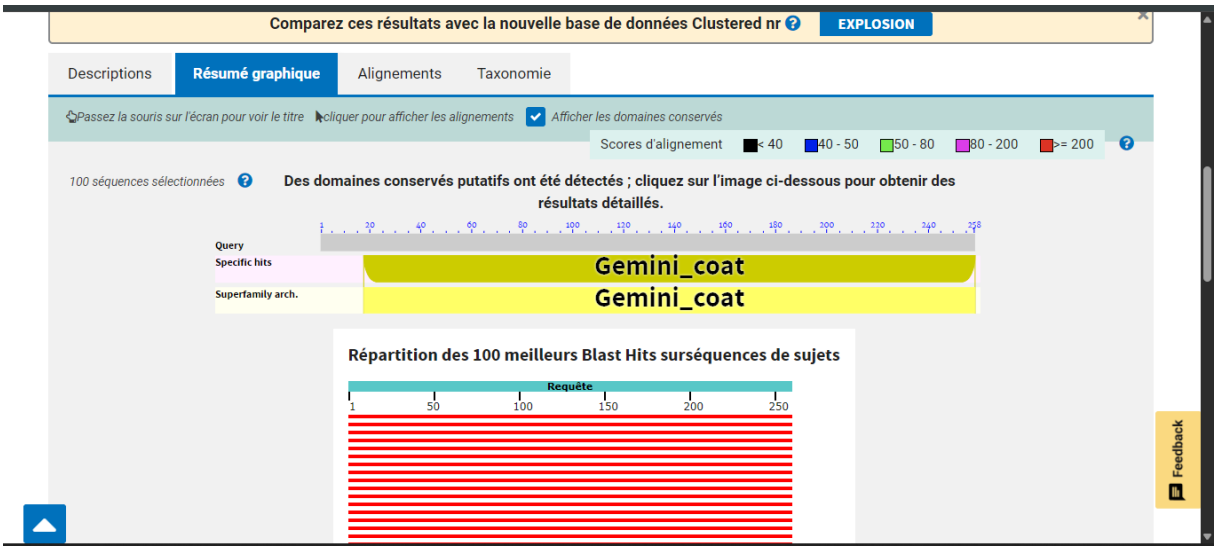


Figure 2: BLASTp graphical summary (RID CX49UVJJ014)

3-2 Top hits

Table 2 lists the top hits. The first entry is the query sequence, which was recovered with 100% identity. The next 14 hits have an E-value of 0.0, 100% coverage, and a length of 258 amino acids. Their identity ranges from 99.61% to 97.29%.

Table 2: Top hits from the BLASTp search with P03561.

Accession	Description	**Score**	**Coverage**	**E-value**	**Identity**
P03561.1	ACMV, West Kenyan 844 (petition)	542	100 %	0,0	100 %
AEO17909.1	*Begomovirus manihotis*	541	100 %	0,0	99,61 %
AEO17902.1	*Begomovirus manihotis*	538	100 %	0,0	99,22 %
AIP88208.1	*Begomovirus manihotis*	536	100 %	0,0	98,84 %
AAF42742.1	CVMA-Uganda Severe	535	100 %	0,0	98,45 %
AAF42735.1	ACMV-Uganda Mild	533	100 %	0,0	98,06 %
UFP62788.1	*Begomovirus manihotis* (AV1)	533	100 %	0,0	97,67 %
CAD20824.1	ACMV-Nigeria-Ogo	533	100 %	0,0	97,29 %

3-3 Alignment with the isolate from Nigeria

The alignment between P03561 and CAD20824.1 has a score of 533 bits and an E-value of 0.0 (Figure 3). The identity is 251/258 (97%) and the similarity is 256/258 (99%). There are no

gaps. Seven positions differ: 46, 78, 123, 126, 137, 189, and 193. The first 45 amino acids are identical.

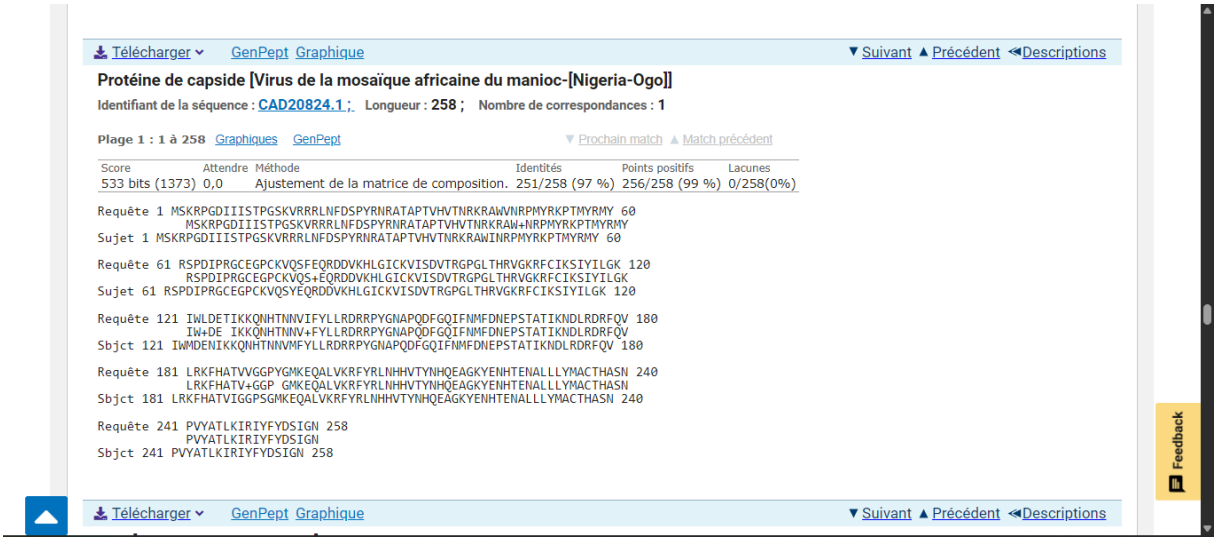


Figure 3: Pairwise alignment between P03561 (query) and CAD20824.1 (ACMV, Nigeria-Ogo).

4-Interpretation

All hits in Table 2 have an E-value of 0.0 and cover the entire protein; they are therefore true homologs. Their identity is very high (97% to 99.6%). The capsid protein varies very little from one isolate to another.

The capsid forms the virus's shell and protects its DNA(Hipp *et al.,* 2017). It also plays a role in transmission by whiteflies. A significant change could potentially render it non-functional. The conserved regions maintain the capsid's structure, while the variable regions are linked to the selection of the vector and the host (Bennett *et al.,* 2024).

BLAST identified the "Gemini_coat" domain characteristic of geminivirus capsids. This confirms the protein's function, even without knowing its specific name.

The isolate from Western Kenya (East Africa) is very closely related to the isolate from Nigeria, with 97% identity and 99% similarity. It is also similar to isolates from Uganda. Consequently, this protein shows little variation across Africa.

Of the 7 differences compared to the Nigerian isolate, 5 involve the replacement of an amino acid with another that has similar properties; these likely have minimal impact on the protein. Only the residues at positions 126 and 193 show more significant changes.

Limitations: BLAST measures similarity, not function. Databases primarily contain isolates that have already been sequenced; therefore, this list does not represent an exhaustive sample of what exists in nature.

Bibliographic references

Bennett, A., J. A. Hull, M. Mietzsch, N. Bhattacharya, P. Chipman and R. McKenna (2024). "Maize streak virus: Single and gemini capsid architecture." Viruses **16**(12): 1861.

Chikoti, P. C. and M. Tembo (2022). "Expansion and impact of cassava brown streak and cassava mosaic diseases in Africa: A review." Frontiers in Sustainable Food Systems **6**: 1076364.

Hipp, K., C. Grimm, H. Jeske and B. Böttcher (2017). "Near-atomic resolution structure of a plant geminivirus determined by electron cryomicroscopy." Structure **25**(8): 1303-1309. e1303.